

Quantum Computing Crash Course

DConf'26

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The Silq programming language

A high-level programming language for quantum computing.

Evolution:

- 2010: Custom D compiler frontend as a hobby project
- 2015: Forked D compiler frontend for the PSI language
- 2018: Forked PSI to create Silq

Silq: Synopsis

```
def bernstein_vazirani[n:!N](const f:  $\mathbb{B}^n \rightarrow$  lifted  $\mathbb{B}$ ){  
  xs := [H(0) for _ in 0..n];  
  if f(xs) { phase( $\pi$ ); }  
  return measure([H(x) for x in xs]);  
}  
  
def f(const xs: $\mathbb{B}^5$ ){  
  bs := [0,1,0,1,1];  
  r := 0;  
  for i in 0..5{  
    r  $\oplus$ = bs[i]·xs[i];  
  }  
  return r;  
}
```

D vs Silq (selection)

Construct	D	Silq
Concatenation	<code>a ~ b</code>	<code>a ~ b</code>
Power	<code>a ^^ b</code>	<code>a ^ b</code>
Xor	<code>a ^ b</code>	<code>a xor b</code> / <code>a ∨ b</code>
Bitwise Xor	<code>a ^ b</code>	<code>a xorb b</code> / <code>a ⊕ b</code>
Declaration	<code>auto a = b</code>	<code>a := b</code>
Ternary expr.	<code>c ? a : b</code>	<code>if a then b else c</code>

D vs Silq types

Construct	D	Silq
Boolean	<code>bool</code>	<code>!B</code> / <code>!ℬ</code>
8-bit unsigned integer	<code>ubyte</code>	<code>!uint[8]</code>
32-bit signed integer	<code>int</code>	<code>!int[32]</code>
Tuple type	<code>Tuple!(S, T, U)</code>	<code>S × T × U</code> / <code>S × T × U</code>
Fixed-length array	<code>T[n]</code>	<code>Tⁿ</code> = <code>T × T × ... × T</code>
Dependent-length array	No	<code>Tⁿ</code> (<code>n</code> a runtime expression)
Dynamic-length array	<code>T[]</code> (slice)	<code>T[]</code> (value type)
Effects	e.g. <code>pure</code>	<code>qfree</code> / <code>mfree</code> / <code>wild</code>
Closures	<code>B delegate(A)</code>	<code>a -> b</code> / <code>a → b</code>
Closures with effects	<code>B delegate(A)pure</code>	<code>A ->qfree B</code> / <code>A →qfree B</code>

NB: Initialization

```
float f; // initialize?  
if(x){  
    f = 2.5f;  
}else{  
    f = 3.1f;  
}  
return f;
```

```
// no initializer needed  
if x {  
    f := 2.5;  
}else{  
    f := 3.1;  
}  
return f;
```

Wall of Shame

Issues for which I have criticized D that Silq still has.

Construct	D	Silq
Effect polymorphism	No	Not yet
Compositional type checking	No	Not yet

Silq · no-effect-polymorphism

```
def map[a,b](const f: a→b)[n:! $\mathbb{N}$ ](xs:a^n) ⇒ [f(x) for x in xs];  
def foo()mfree ⇒ map(H)(vector(3,0)); // error
```

Silq · non-compositional

```
def bar(){  
  x := 0:ℬ; // ok  
  y := H(dup(x));  
  return y;  
}  
  
def bar2(){  
  x := 0:ℬ;  
  y := H(dup(x));  
  return (x,y);  
}  
  
def baz(){  
  (x,y) := bar2(); // error  
  return y;  
}
```

Quantum: States

Classical state: $\{x \mapsto 1; y \mapsto 0\}$

Standard basis state: $|x \mapsto 1; y \mapsto 0\rangle$

More compact notation: $|1\rangle_x |0\rangle_y$

Quantum: Amplitudes and Measurements

$$|+\rangle = \sqrt{0.5} \cdot |0\rangle + \sqrt{0.5} \cdot |1\rangle$$

Silq · hadamard-demo

```
def main(){  
  x := H(0);  
  return x;  
}
```

Quantum: Reversibility

Quantum operations other than measurements are reversible.

Silq · non-reversible

```
def f(x:int[2]){  
  y := x·x-1;  
  return y;  
}  
  
def main() ⇒ f(2:!int[2]);
```

Quantum: Linearity

Measurement-free operations U apply like $U(\alpha \cdot |\varphi\rangle + \beta \cdot |\psi\rangle) = \alpha \cdot U|\varphi\rangle + \beta \cdot U|\psi\rangle$

Silq · quantum-parallelism

```
def main(){  
  x := H(0);  
  y := if x then 3:uint[8] else 16:uint[8];  
  // |x,y> = √0.5 · |0,16> + √0.5 · |1,3>  
  y += 3;  
  return (x,y);  
}
```

Quantum: Unitary Operations

- More generally, measurement-free operations preserve the overall probability (sum of squared amplitude magnitudes).
- This implies that they are “unitary”: high-dimensional rotations (and reflections).

Quantum: Phases

In addition to a magnitude, amplitudes have a direction.

Silq · phase-demo

```
def main() {  
  x := H(0);  
  if x { phase( $\pi/10$ ); }  
  return x;  
}
```

Quantum: Interference

Example: Hadamard operator

$$H |0\rangle = |+\rangle = \sqrt{0.5} \cdot |0\rangle + \sqrt{0.5} \cdot |1\rangle$$

$$H |1\rangle = |-\rangle = \sqrt{0.5} \cdot |0\rangle - \sqrt{0.5} \cdot |1\rangle$$

Linearity holds generally, so we get interference:

$$\begin{aligned} H |+\rangle &= \sqrt{0.5} \cdot H |0\rangle + \sqrt{0.5} \cdot H |1\rangle \\ &= \sqrt{0.5} \cdot |+\rangle + \sqrt{0.5} \cdot |-\rangle \\ &= 0.5 \cdot |0\rangle + 0.5 \cdot |1\rangle + 0.5 \cdot |0\rangle - 0.5 \cdot |1\rangle \\ &= |0\rangle. \end{aligned}$$

Similarly, $H |-\rangle = |1\rangle$.

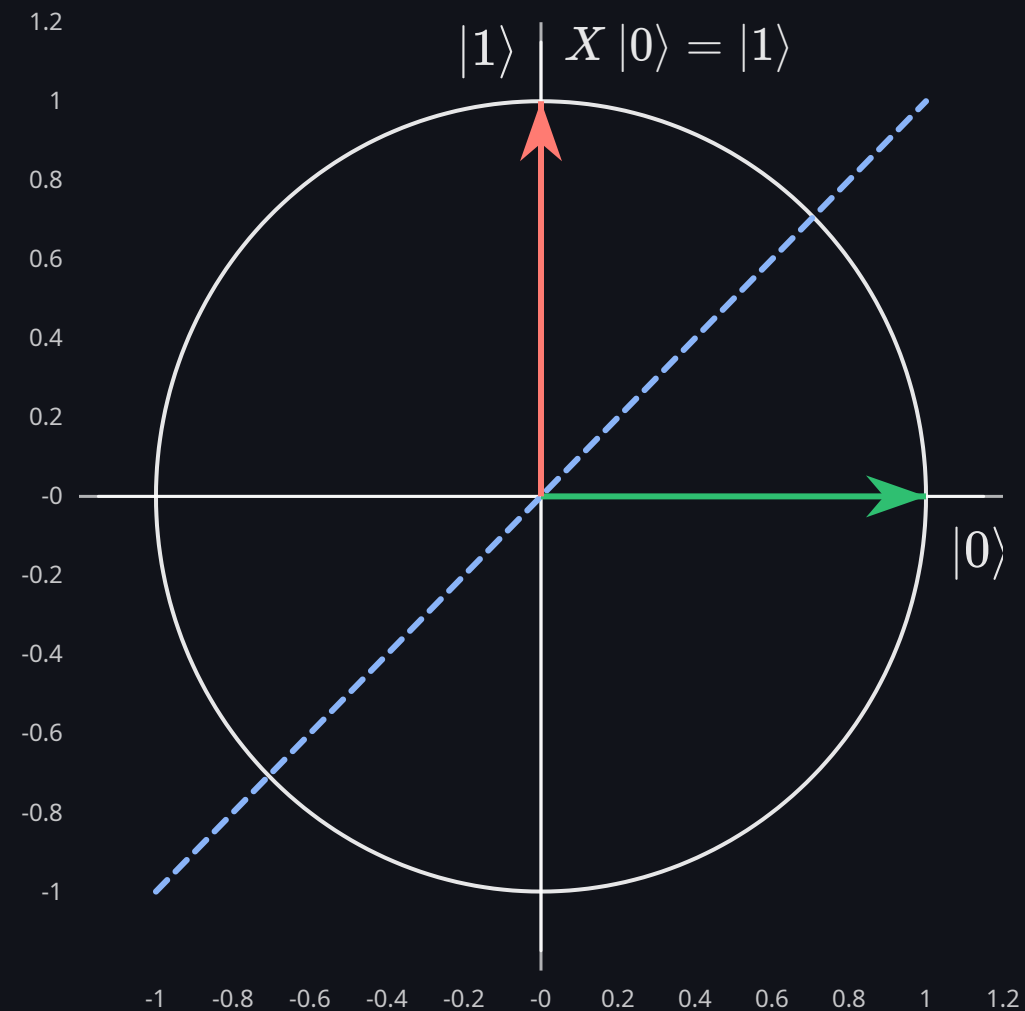
Hadamard demo

Silq · hadamard-demo

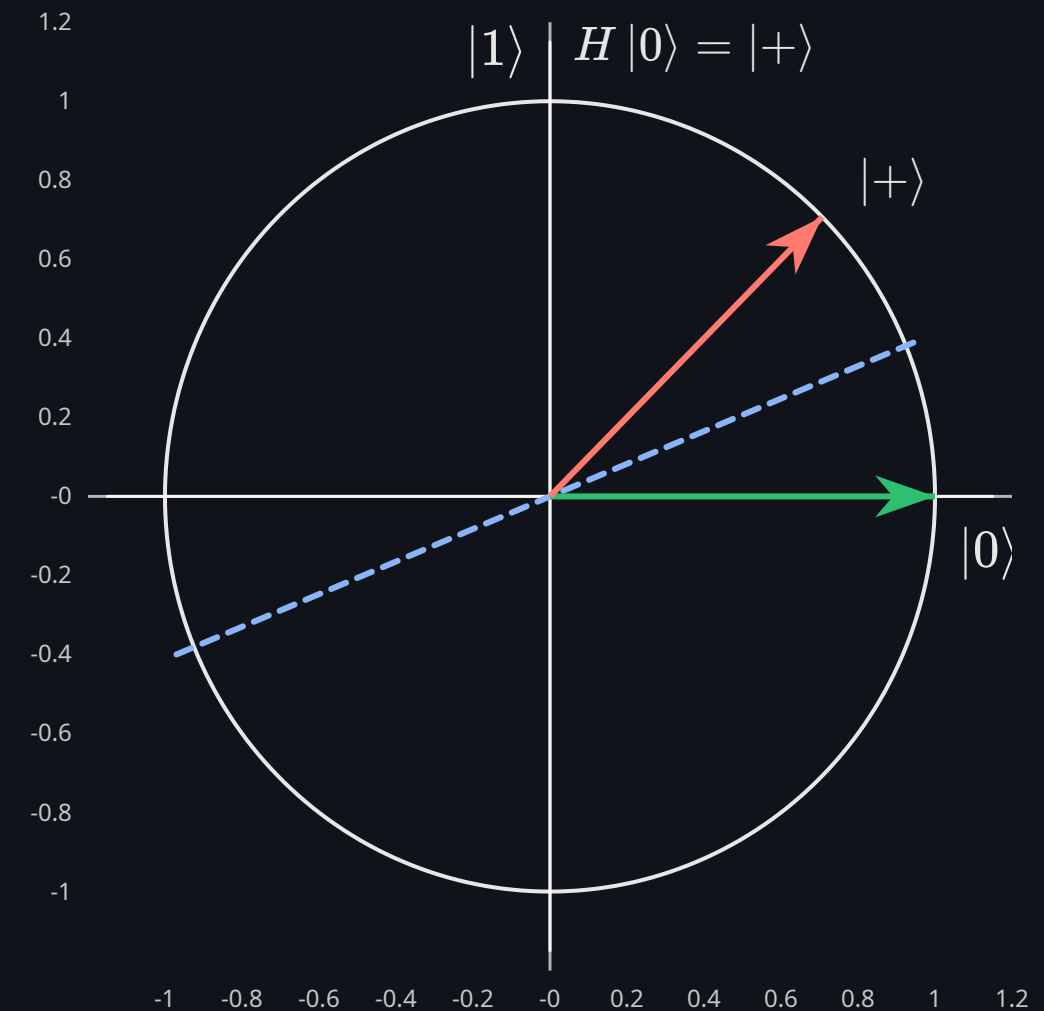
```
def main(){  
  x := H(0);  
  return x;  
}
```

Negation and H as Reflections

X gate



H gate



Phase Estimation

Silq · phase-estimation

```
t := 6;
def phase_u(const x:!uint[t]){
  phase(2·π·(x as !ℕ)/2^t);
}

def f(){
  x := 42:!uint[t];
  phase_u(x);
}

def estimate_phase(const f: 1 →mfree 1){
  return x; // TODO
}

def main() ⇒ estimate_phase(f);
```

Application: Factoring Semiprimes (Shor's Algorithm)

Input: $N = p \cdot q$, for primes p, q

Output: p and q .

Classical Observations

It suffices to find $2 \leq a < N$ and even r with $a^r \% N = 1$, $a^{(r/2)} \% N$ is not 1 or $N - 1$.

Then: $0 = (a^r - 1) \% N = ((a^{(r/2)} + 1) \% N * (a^{(r/2)} - 1) \% N) \% N$

Therefore $\gcd(a^{(r/2)} + 1, N)$ is a nontrivial factor of N .

Quantum Observations (1)

$$\begin{aligned} |\varphi^{(i)}\rangle = & \sqrt{1/r}(\angle(-2 \cdot \pi \cdot 0 \cdot i/r) \cdot |a^0\rangle + \\ & \angle(-2 \cdot \pi \cdot 1 \cdot i/r) \cdot |a^1\rangle + \\ & \angle(-2 \cdot \pi \cdot 2 \cdot i/r) \cdot |a^2\rangle + \\ & \dots + \\ & \angle(-2 \cdot \pi \cdot (r-1) \cdot i/r) \cdot |a^{r-1}\rangle) \end{aligned}$$

$$|1\rangle = \sqrt{1/r}(|\varphi^{(0)}\rangle + |\varphi^{(1)}\rangle + |\varphi^{(2)}\rangle + \dots + |\varphi^{(r-1)}\rangle)$$

$$\begin{aligned}
|\varphi^{(0)}\rangle &= \sqrt{1/8} (\rightarrow |a^0\rangle + \rightarrow |a^1\rangle + \rightarrow |a^2\rangle + \rightarrow |a^3\rangle + \rightarrow |a^4\rangle + \rightarrow |a^5\rangle + \rightarrow |a^6\rangle + \rightarrow |a^7\rangle) \\
|\varphi^{(1)}\rangle &= \sqrt{1/8} (\rightarrow |a^0\rangle + \searrow |a^1\rangle + \downarrow |a^2\rangle + \swarrow |a^3\rangle + \leftarrow |a^4\rangle + \nwarrow |a^5\rangle + \uparrow |a^6\rangle + \nearrow |a^7\rangle) \\
|\varphi^{(2)}\rangle &= \sqrt{1/8} (\rightarrow |a^0\rangle + \downarrow |a^1\rangle + \leftarrow |a^2\rangle + \uparrow |a^3\rangle + \rightarrow |a^4\rangle + \downarrow |a^5\rangle + \leftarrow |a^6\rangle + \uparrow |a^7\rangle) \\
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|\varphi^{(5)}\rangle &= \sqrt{1/8} (\rightarrow |a^0\rangle + \nwarrow |a^1\rangle + \downarrow |a^2\rangle + \nearrow |a^3\rangle + \leftarrow |a^4\rangle + \searrow |a^5\rangle + \uparrow |a^6\rangle + \swarrow |a^7\rangle) \\
|\varphi^{(6)}\rangle &= \sqrt{1/8} (\rightarrow |a^0\rangle + \uparrow |a^1\rangle + \leftarrow |a^2\rangle + \downarrow |a^3\rangle + \rightarrow |a^4\rangle + \uparrow |a^5\rangle + \leftarrow |a^6\rangle + \downarrow |a^7\rangle) \\
|\varphi^{(7)}\rangle &= \sqrt{1/8} (\rightarrow |a^0\rangle + \nearrow |a^1\rangle + \uparrow |a^2\rangle + \nwarrow |a^3\rangle + \leftarrow |a^4\rangle + \swarrow |a^5\rangle + \downarrow |a^6\rangle + \searrow |a^7\rangle)
\end{aligned}$$

Quantum Observations (2)

$$U_f |x\rangle = |f(x)\rangle := |a \cdot x \% N\rangle .$$

$$\begin{aligned} U_f |\varphi^{(i)}\rangle = & \sqrt{1/r} (\angle(-2 \cdot \pi \cdot 0 \cdot i/r) \cdot U_f |a^0\rangle + \\ & \angle(-2 \cdot \pi \cdot 1 \cdot i/r) \cdot U_f |a^1\rangle + \\ & \angle(-2 \cdot \pi \cdot 2 \cdot i/r) \cdot U_f |a^2\rangle + \\ & \dots + \\ & \angle(-2 \cdot \pi \cdot (r-1) \cdot i/r) \cdot U_f |a^{r-1}\rangle) \end{aligned}$$

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Quantum Observations (2)

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Quantum Observations (2)

$$U_f |x\rangle = |f(x)\rangle := |a \cdot x \% N\rangle .$$

$$\begin{aligned} U_f |\varphi^{(i)}\rangle = & \angle(2 \cdot \pi \cdot i / r) \sqrt{1/r} (\angle(-2 \cdot \pi \cdot 0 \cdot i / r) \cdot |a^0\rangle + \\ & \angle(-2 \cdot \pi \cdot 1 \cdot i / r) \cdot |a^1\rangle + \\ & \angle(-2 \cdot \pi \cdot 2 \cdot i / r) \cdot |a^2\rangle + \\ & \dots \\ & \angle(-2 \cdot \pi \cdot (r - 1) \cdot i / r) \cdot |a^{r-1}\rangle) \end{aligned}$$

Quantum Observations (2)

$$U_f |x\rangle = |f(x)\rangle := |a \cdot x \% N\rangle .$$

$$U_f |\varphi^{(i)}\rangle = \angle(2 \cdot \pi \cdot i / r) \cdot |\varphi^{(i)}\rangle$$

$$\begin{aligned}
|\varphi^{(0)}\rangle &= \sqrt{1/8} (\rightarrow |a^0\rangle + \rightarrow |a^1\rangle + \rightarrow |a^2\rangle + \rightarrow |a^3\rangle + \rightarrow |a^4\rangle + \rightarrow |a^5\rangle + \rightarrow |a^6\rangle + \rightarrow |a^7\rangle) \\
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\end{aligned}$$

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\end{aligned}$$

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\downarrow |\varphi^{(6)}\rangle &= \sqrt{1/8} (\downarrow |a^0\rangle + \rightarrow |a^1\rangle + \uparrow |a^2\rangle + \leftarrow |a^3\rangle + \downarrow |a^4\rangle + \rightarrow |a^5\rangle + \uparrow |a^6\rangle + \leftarrow |a^7\rangle) \\
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\end{aligned}$$

Quantum Observations (3)

$$U_f |x\rangle = |f(x)\rangle := |a \cdot x \% N\rangle .$$

$$U_f |\varphi^{(i)}\rangle = \angle(2 \cdot \pi \cdot i/r) \cdot |\varphi^{(i)}\rangle$$

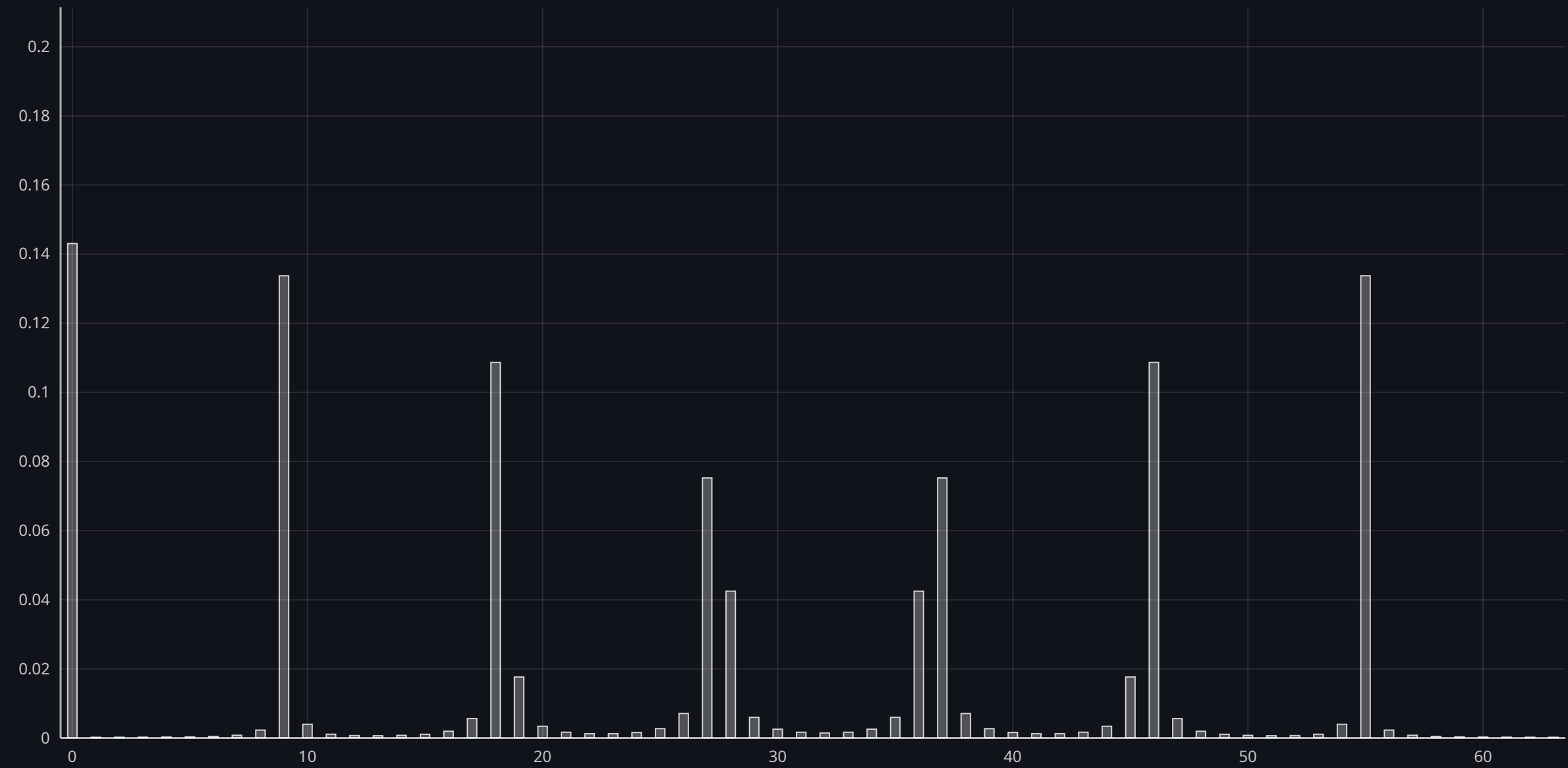
t -bit phase estimation on `f` on state $|i\rangle$ yields (roughly):

$$\text{ep}(U_f) |\varphi^{(i)}\rangle = |2^t \cdot i/r\rangle \cdot |\varphi^{(i)}\rangle$$

Quantum Observations (4)

$$\begin{aligned} \text{ep}(U_f) |1\rangle = & \sqrt{1/r} (|2^t \cdot 0/r\rangle \cdot |\varphi^{(0)}\rangle + \\ & |2^t \cdot 1/r\rangle \cdot |\varphi^{(1)}\rangle + \\ & |2^t \cdot 2/r\rangle \cdot |\varphi^{(2)}\rangle + \\ & \dots + \\ & |2^t \cdot (r-1)/r\rangle \cdot |\varphi^{(r-1)}\rangle) \end{aligned}$$

Roughly: If we measure this, we can randomly sample an integer approximation of $2^t \cdot i/r$ for some i . If $\gcd(i, r) = 1$ (quite likely), we may be able to recover r .



r: 7

t: 6

Phase Estimation

Silq · phase-estimation

```
t := 6;
def phase_u(const x:!uint[t]){
  phase(2·π·(x as !ℕ)/2^t);
}

def f(){
  x := 42:!uint[t];
  phase_u(x);
}

def estimate_phase(const f: 1 →mfree 1){
  return x; // TODO
}

def main() ⇒ estimate_phase(f);
```

Implementation: Shor's Algorithm

Silq · shor

```
def main(){
  N:=21;
  k:=shor(N);
  assert(1<k&&k<N&& N%k=0);
  return k;
}

def shor(N:!N){
  t := 2·ceil(log2(N))+1 coerce !N;
  while true {
    a := 2+uniform(n sub 2);
    r_tilde := ...; // TODO
    for r in candidates(r_tilde/2^t) {
      k := gcd(a^(r div 2)+1 as !N,n);
      if 1<k && k<n { return k; }
    }
  }
}

def candidates(q:!Q): !N[] {
  r := [];
  (b, d) := (1, 0);
  while a!=0 {
```

Questions?